



National Laboratory of the Rockies

**National Laboratory of the Rockies
Energy Systems Integration Facility
High Performance Computing
Request for Information
RFI-2026029603**

**Date of Issuance: May 5, 2026
Response Deadline: 6/17/2026, 5:00 PM MST**

The National Laboratory of the Rockies (NLR) is the U.S. Department of Energy's primary national laboratory for energy systems research and development. As the energy systems laboratory, NLR's unique strength lies in developing and integrating a broad array of energy technologies into robust, resilient systems, bridging foundational research with practical applications to lower energy costs, drive economic growth, bolster national security, and deliver abundant and reliable energy.

The Energy Systems Integration Facility (ESIF) at NLR hosts Kestrel, the primary high-performance computing system for work supported by the Department of Energy's Office of Critical Minerals and Energy Innovation (CMEI). NLR is preparing to replace Kestrel with a system that will provide the performance and capacity required to meet the mission needs of CMEI in the 2028–2032 timeframe. The replacement system (ESIF-HPC-4) must be compatible with the ESIF data center, which balances energy efficiency equally with computational performance. The purpose of this request for information (RFI) is to gain further understanding of the currently available technologies in the marketplace. The ESIF-HPC-4 delivery schedule must be coordinated with available power infrastructure upgrades, with substantial first-phase system delivery of the ESIF-HPC-4 anticipated in December 2027, with a potential second-phase delivery in May 2029. NLR anticipates the following schedule:

- June 2026 – RFI Due Date
- November 2026 – Anticipated Request for Proposal (RFP) Release Date
- January 2027 – Anticipated RFP Proposal Due Date
- June 2027 – Anticipated Contract Award
- December 2027 – Anticipated Phase 1 Delivery
- May 2028 – Anticipated Phase 1 System Acceptance and Production-Ready Determination
- May 2029 – Potential Phase 2 Delivery.
- December 2029 – Anticipated Phase 2 System Acceptance and Production-Ready Determination

Context

Desired System Characteristics

NLR seeks a system for ESIF-HPC-4 composed of the following:

1. Graphics Processing Unit (GPU)-accelerated compute: The majority of compute cycles will come from GPU-based nodes. These nodes must support accelerated computing and Artificial Intelligence/ Machine Learning (AI/ML) workloads and deliver strong performance for traditional HPC applications that are increasingly GPU-enabled.
2. Central Processing Unit (CPU)-based compute: A smaller portion of the workloads (anticipated at <15% of total compute cycles) requires CPU-based nodes. These nodes must provide high per-core performance for supporting efficient workloads that are not GPU-accelerated. The CPU node architecture will complement the GPU-accelerated portion of the system, as a part of a single, integrated system for enabling consistent build environment and management across the entire system.
3. Storage: The system will include the following storage characteristics:
 - a. Performance tier (i.e., parallel file system) providing high-bandwidth, low-latency support for large-scale HPC workloads.
 - b. Capacity tier (i.e., object store) providing high-durability, cost-efficient, scalable storage optimized for large, shareable datasets, small random reads, and long tails of access for a growing AI workload.
4. High-speed interconnect: The system will have a low-latency, high bandwidth interconnect to support scalability of multi-node applications running across a majority of the system, as well as to maintain a balance with node-level computational performance.
5. Consumption-based compute augmentation: The system may include an option to access consumption-based compute resources, provided either on-premise or off-premise, to flexibly augment capacity for all or part of the workload. While this capability is anticipated to represent a small fraction of total compute cycles, it will address critical, time-sensitive computing needs that arise at various points in an allocation cycle.

Summary of Anticipated Workload

NLR's systems have historically supported a heterogeneous mix of traditional modeling and simulation workloads, as well as high-throughput ensembles of smaller jobs. AI/ML-driven workloads are increasingly becoming a larger percentage of compute workloads, as are hybrid and data intensive workloads. Specifically:

1. Artificial Intelligence: AI and machine learning (AI/ML) is a growing workload on NLR systems. NLR anticipates a broad range of use cases as researchers and application developers explore and integrate AI/ML technologies into their workflows. These workloads are comprised primarily of training and inference jobs ranging from single-node to multi-node scale requiring high memory bandwidth and GPU-GPU communication both between and within nodes. Large language models (LLMs) and generative AI workloads are also

experiencing growth. Many such workloads at NLR will couple AI/ML with modeling and simulation codes, requiring double-precision data types and fast data movement between CPUs, GPUs and storage. These AI/ML-powered workflows may require an optimized storage layer that can handle high Input/Output (I/O), i.e., frequent reads of large numbers of training files and model artifact writes.

2. **Modeling and Simulation:** The modeling and simulation workload is characterized by large-scale, tightly coupled jobs spanning hundreds of nodes with both strong and weak scaling requirements and requiring high-precision data types (i.e. FP64). The largest runs scale to ~50% of the system, though the majority typically consume 5-10% of compute resources per job. In addition, this category includes high throughput ensembles of smaller jobs using fewer than 100 nodes, but many jobs are often single node only. Simulation codes are increasingly migrating to accelerated paradigms, and many require scaling to multiple GPU nodes. A minority of the modeling and simulation workloads will continue to require CPU-only nodes. The interconnect must provide sufficient support for all modeling and simulation workloads, provide application scalability, and balance node-level computation performance.
3. **Hybrid and Data Intensive:** This category of workload includes digital twins, distributed data analytics, visualization, experimental interoperability, and edge compute capabilities, often requiring container-based deployments or orchestrated workflows. Some use cases require real-time on-demand access to compute resources rather than the traditional HPC queue, to enable operational digital twin research, AI-controlled experimental facilities, and hardware-in-the-loop experiments.

What information are we seeking?

NLR requests that organizations submitting a response to this RFI include the following information in their response:

1. Feedback on the draft Technical Specifications Attachment 1, draft Benchmark Instructions, and Benchmark Reporting at <https://github.com/NatLabRockies/ESIFHPC4/> and the anticipated timeline (see applicable attachments and links section below), including any areas where your organization's planned product roadmap may not align with the technical requirements or schedule outlined in this RFI.
2. The date your organization anticipates that a fully executed subcontract would be required to deliver the system by the anticipated date of December 2027.
3. Large-scale HPC procurements may involve extended delivery, installation and acceptance testing periods, potentially up to or exceeding one year prior to final system acceptance. Please describe payment structures that would support your organization's ability to successfully deliver a system of this scale. Specifically, please address your organization's ability to deliver the HPC system described in this RFI under a (i) lease-to-own payment model with final payment upon full system acceptance; (ii) milestone-based payment model(s) (e.g., payments tied to delivery, installation, or interim acceptance milestones, with final payment upon full acceptance); or (iii) other payment models that would enable your organization's successful delivery of this system. *This RFI is issued solely for information-gathering purposes and does not represent a commitment to, or guarantee of, a specific payment model in any future procurement.*

4. A description of your organization's ability to provide early access to a subset of system compute components, such as pre-production hardware and software environments, ahead of full system delivery.
5. Information on your organization's HPC system cooling functionality. Specifically, NLR operates a warm-water cooled HPC data center. NLR will provide cooling water to the system that can be as warm as twenty-three point eight degrees Celsius (23.8 °C). Please state if, under normal compute loads, your organization's HPC system would return water that is (i) thirty-seven degrees Celsius (37 °C) or warmer and is (ii) suitable for waste heat capture and reuse.
6. NLR prefers 480V three phase power feeds up to 150A for compute intensive nodes. If your organization's HPC system prefers a different voltage or you anticipate a direct current (DC) system or other power distribution approach, please describe the power characteristics of system configurations on your product roadmap that would be available within the planned ESIF HPC-4 timeframe.
7. NLR prefers full Direct Liquid Cooling (DLC) for the compute intensive nodes. If your HPC system utilizes a hybrid cooling architecture (i.e., where CPU or GPU components are DLC and other components are air cooled) please describe the system cooling architecture. Also describe the typical percentage of DLC versus air cooling for a rack or collection of racks for your system.
8. NLR is interested in emerging and first-of-a-kind power and cooling technologies that may be available within the HPC-4 timeframe (e.g., 800V direct current (DC) distribution, two-phase or immersion cooling, or other advanced architectures). Please describe any such capabilities on your product roadmap, including expected availability, system integration considerations, and any potential advantages in cost, schedule, efficiency, or performance. Additionally, indicate whether your organization would be interested in partnering on a demonstration deployment (for part or all of the system), including opportunities for cost sharing or co-development.
9. If applicable, feedback on technical feasibility, availability, and suitability of consumption-based cloud compute models for all or portions of the anticipated workload.
10. If applicable, a summary of your organization's cost model for on-demand/consumption computing. Please explain how all costs are forecasted, monitored, and controlled under this model. In addition, describe options to procure defined blocks of capacity at a fixed price, with the ability to add additional blocks at the fixed price over a multi-year contract. Highlight any constraints, dependencies, or commercial practices that would impact your organization's ability to offer this procurement structure.
11. NLR has some workloads that require storage (RAM/Disk) in excess of what can typically be placed onboard of a dense water-cooled rack solution. Please describe if, in the anticipated timeline for this system, a Compute Express Link (CXL)-connected tier or other technology in a standard rack and node deployment for a subset of nodes is an option (as opposed to purchasing non-standard nodes).

Applicable attachments and links:

****Note: all documents are in draft form and subject to change****

Technical Specifications (Attachment 1), is posted as an attachment in draft form to this RFI. For draft benchmark instructions, and benchmark reporting refer to the instructions and

documentation available on our GitHub code repository here:

<https://github.com/NatLabRockies/ESIFHPC4/>

THIS IS NOT A REQUEST FOR PROPOSAL. This request is a market research effort. Your submittal is not in response to a Request for Proposals (RFP). During this RFI period, NLR will not participate in informational meetings or technology demonstrations by vendors. Any costs associated with the development of a response to this request will not be reimbursed by NLR, and submission does not guarantee or otherwise promise inclusion in any forthcoming RFP. In the event this RFI results in a solicitation, NLR reserves the right to set aside in total or in part any resultant award.

Responses to this RFI should be submitted in the form of either an Adobe Acrobat PDF document or a Microsoft Word document attached to an electronic mail message addressed to Chris Sloneker at ASP.Responses@nlr.gov with the subject line of RFI-2026029603. Attachment filenames should contain RFI-2026029603 and the name of the company or organization providing the responses. Questions and responses to this RFI will be accepted through June 27th 2026 5:00 PM MST.